“Towards an Approach for Improving Exploratory Testing Tour Assignment based on Testers’ Profile” DOI: 10.5220/0011113800003179
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This paper presents an empirical study examining the relationship between a software tester's profile (knowledge, expertise, education) and their effectiveness and preference when applying "tours" within the Tourist Metaphor for exploratory software testing. The goal is to inform the development of a recommendation system to improve the assignment of test tasks.

The research employed an empirical approach, involving 60 participants with varying backgrounds—40 undergraduate students and 20 professionals enrolled in specialized courses. In the data collection phase, the research procedures employed were: documentary research; bibliographic research; and action research. A questionnaire was applied to each participant, in order to identify their characteristics. The participants then engaged in an exploratory testing activity using 16 distinct tours within the Tourist Metaphor framework applied to a simulated service, and their preferences and performance (measured by the ratio of defects found to the number of test cases executed) were analyzed to identify patterns and correlations between profile attributes and testing behavior.

The study found that tester profiles significantly influenced both the choice of tours and the efficiency in defect identification. Tours such as the "Antisocial Tour," "Couch Potato," and "Obsessive-Compulsive Tour" were the most frequently chosen, especially by testers with more experience, indicating a preference for tours that either challenge expected behaviors or emphasize repeat actions. However, choices were also driven by perceived execution time, suggesting a tension between efficiency and perceived ease. Importantly, testers with higher educational backgrounds tended to prefer tours that explored edge cases or imposed greater testing loads, supporting the notion that deeper expertise influences testing strategy.

These findings reinforce the value of considering individual tester characteristics in task allocation, pointing to the potential for more "humanized" and adaptive software testing strategies. For researchers, the study opens pathways for further investigation into the use of artificial intelligence in building intelligent recommendation systems that match test tasks with tester profiles. For practitioners, this work suggests that understanding the cognitive and experiential dimensions of testers can improve test planning and execution outcomes, ultimately enhancing productivity and software quality. The envisioned system could transform manual testing from a static process into a dynamic and personalized activity that capitalizes on human strengths.